

Safety Data Sheet

according to 1907/2006/EC, Article 31

Printing date 01.09.2015 Version: 1 Revision: 01.09.2015



1. Identification of the substance/preparation and of the company/undertaking

1.1 Product identifier

Trade name: PU INK_CY

1.2 Relevant identified uses of the substance or mixture and uses advised against

Material of Use: Industrial applications: Inkjet ink for drop-on-demand digital printing process.

1.3 SDS No.

1.4 Details of the supplier of the safety data sheet

Manufacturer/Supplier: Jetbest Corporation

Address: No. 402-1, Sec. 4, Chun Hwa Road, Hsinchu, 300, Taiwan

Phone: +886-530-5888

e-mail: sales@jetbest.com

Only Representative

Address: Instituto Suizo para el Fomento de la Seguridad-Swissi España S.L.U Calle Lope de Vega 30 Entlo. E-08005 Barcelona

e-mail: reach@swissiesp.com

Emergency telephone number:

Manufacturer: +886-913-323-577 (24hours)

2. Hazards identification

2.1 Classification of the substance or mixture

Classification according to Regulation (EC) No 1272/2008 [CLP]



Danger

Acute Tox. 4 H302 Harmful if swallowed.

Eye Irrit. 2 H318 Causes serious eye irritation.

Skin Irrit.2 ; H315 Causes skin irritation.

STOT SE 3: H336 May cause drowsiness or dizziness.

2.2 Label elements

Labelling according to Regulation (EC) No 1272/2008 [CLP]

Hazard pictograms



Signal word: Danger

Hazard statements:

H302 Harmful if swallowed.

H315 Causes skin irritation.

H318 Causes serious eye irritation.

H336 May cause drowsiness or dizziness.

Precautionary statements:

P261 Avoid breathing dust/ fume/ gas/ mist/ vapours/ spray.

P280 Wear protective gloves/ eye protection/ face protection.

P302+P352 - IF ON SKIN: Wash with plenty of soap and water.

P321 - Specific treatment (see first aid statements on this label).

P305 + P351 + P338 IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.

P362 - Take off contaminated clothing and wash before reuse.

2.3 Other hazards

Results of PBT and vPvB assessment

PBT: Not applicable.

vPvB: Not applicable.

3. Composition/information on ingredients

Chemical characterization: Mixture
Ink Jet printing ink in organic solvents.

Ingredients	CAS-No.	EINECS	EU Pre-registration No.	Percent	Classification	
					67/548/EEC	Regulation (EC) No. 1272/2008 [CLP]
2-Ethoxy Ethyl Ether	112-36-7	203-963-7	Not available for the moment	<25%	Xi; R38	Skin Irrit.2 H315
Diethylene Glycol Ethyl Methyl Ether	1002-67-1	213-690-5	Not available for the moment	40%–70%	—	—
Gamma-Butyrolactone	96-48-0	202-509-5	Not available for the moment	10%–20%	Xi; R22, R41; R67	Acute Tox. 4 H302 Eye Irrit. 2 H318 STOT.SE 3 H336

4. First aid measures

4.1 Description of first aid measures

Eye contact: Rinse thoroughly with plenty of water for at least 15 minutes and consult a physician.
Inhalation: If breathed in, move person into fresh air. If not breathing, give artificial respiration. Consult a physician.
Skin contact: Wash off with soap and plenty of water. Consult a physician.
Ingestion: Do NOT induce vomiting. Never give anything by mouth to an unconscious person. Rinse mouth with water. Consult a physician.

4.2 Most important symptoms and effects, both acute and delayed

Potential acute health effects

Eye contact : No known significant effects or critical hazards.
Inhalation : No known significant effects or critical hazards.
Skin contact : No known significant effects or critical hazards.
Ingestion : No known significant effects or critical hazards.

Over-exposure signs/symptoms

Eye contact : No specific data.
Inhalation : No specific data.
Skin contact : No specific data.
Ingestion : No specific data.

4.3 Indication of any immediate medical attention and special treatment needed

Treat symptomatically.

5. Fire-fighting measures

5.1 Extinguishing media

Suitable extinguishing media

Alcohol-resistant foam, dry chemical, carbon dioxide (CO₂), water-spray.

5.2 Special hazards arising from the substance or mixture

Carbon oxides

5.3 Advice for firefighters

Use breathing apparatus with independent air supply.
Protective suit.

5.4 Further information

Use water spray to cool unopened containers

5.5 NFPA Ratings:

Health: 2 Flammability: 2 Reactivity: 0

Hazard Scale: 0 = Minimal 1 = Slight 2 = Moderate 3 = Serious 4 = Severe

6. Accidental release measures

6.1 Personal precautions, protective equipment and emergency procedures

Use personal protective equipment. Avoid breathing vapors, mist or gas. Ensure adequate ventilation. Remove all sources of ignition. Beware of vapours accumulating to form explosive concentrations. Vapours can accumulate in low areas.

6.2 Environmental precautions

Prevent further leakage or spillage if safe to do so. Do not let product enter drains.

6.3 Methods and materials for containment and cleaning up

Contain spillage, and then collect with an electrically protected vacuum cleaner or by wet-brushing and place in container for disposal according to local regulations (see section 13). Keep in suitable, closed containers for disposal.

6.4 Reference to other sections

For disposal see section 13.

7. Handling and storage

7.1 Handling

Avoid contact with skin and eyes. Avoid inhalation of vapour or mist. Keep away from sources of ignition - No smoking. Take measures to prevent the build up of electrostatic charge.

7.2 Conditions for safe storage, including any incompatibilities

Store in cool place. Keep container tightly closed in a dry and well-ventilated place. Containers which are opened must be carefully resealed and kept upright to prevent leakage.

7.3 Specific end uses

no data available

8. Exposure controls/personal protection

8.1 Control parameters

Components with workplace control parameters

8.2 Exposure controls

Appropriate engineering controls

Handle in accordance with good industrial hygiene and safety practice. Wash hands before breaks and at the end of workday.

Personal protective equipment

Eye/face protection

Face shield and safety glasses Use equipment for eye protection tested and approved under appropriate government standards such as NIOSH (US) or EN 166(EU).

Skin protection

Handle with gloves. Gloves must be inspected prior to use. Use proper glove removal technique (without touching glove's outer surface) to avoid skin contact with this product. Dispose of contaminated gloves after use in accordance with applicable laws and good laboratory practices.

Wash and dry hands.

The selected protective gloves have to satisfy the specifications of EU Directive 89/686/EEC and the standard EN 374 derived from it.

Body Protection

Complete suit protecting against chemicals, The type of protective equipment must be selected according to the concentration and amount of the dangerous substance at the specific workplace.

Respiratory protection

Where risk assessment shows air-purifying respirators are appropriate use a full-face respirator with multi-purpose combination (US) or type ABEK (EN 14387) respirator cartridges as a backup to engineering controls. If the respirator is the sole means of protection, use a full-face supplied air respirator. Use respirators and components tested and approved under appropriate government standards such as NIOSH (US) or CEN (EU).

9. Physical and chemical properties

1	Color	Cyan
2	Odor	Slight odor
3	Boiling point/boiling range of ink	approx. 176 °C or higher
4	Melting Point/Melting Range	No data available
5	Flash point of ink	approx. 70 °C
6	Auto-Ignition Temperature	not below 220 °C
7	Flammability(solid, gas)	Not Applicable
8	Explosive Properties	Lower limits: 2.7 vol% Upper limits: 15.6 vol% (gamma-Butyrolactone)
9	Vapour Pressure	No data available
10	Specific Gravity	0.97 ± 0.01(25°C)
11	Solubility	No data available
12	Water solubility	Easily soluble (Diethylene glycol diethyl ether)
13	Viscosity	4.3 ± 0.5 cps
14	pH	Not applicable
15	Oxidizing properties	No data available
16	Vapor Density	Not Applicable

The Physical and chemical data given in Section 9 are typical values for this product and are not intended to be product specifications.

10. Stability and reactivity

10.1 Reactivity

- no data available
- 10.2 Chemical stability
 - no data available
- 10.3 Possibility of hazardous reactions
 - no data available
- 10.4 Conditions to avoid
 - Heat, flames and sparks.
- 10.5 Incompatible materials
 - Strong oxidizing agents, Strong bases
- 10.6 Hazardous decomposition products
 - Other decomposition products - no data available

11. Toxicological information

11.1 Routes of Overexposure: Eye, skin, inhalation, and oral ingestion

11.2 Health Hazards:

Acute Health Hazards: Overexposure of the eye surface to ink may be mildly irritating. Overexposure of ink contact with the skin may cause irritation and in some people, swelling and redness. Intentional inhalation to ink vapors may result in respiratory tract irritation. Intentional or accidental oral ingestion may cause an upset stomach.

Chronic Health Hazards: No information available

Mutagenicity: No information available

Carcinogenicity: No information available

11.3 Toxicity:

Acute Toxicity Data:

2-Ethoxy Ethyl Ether:

LD50 Oral, rat: 4,970 mg/kg

LD50 Skin, rabbit: 6,700 uL/kg

Draize test, Eye rabbit: 50 mg Moderate

Diethylene Glycol Ethyl Methyl Ether:

LD50 Oral, rat: >6,500mg/kg

LD50 Dermal, rabbit: >7,070mg/kg

Gamma-Butyrolactone:

LD50 Oral, rat: 1,540 mg/kg

LD50 Oral, mouse: 1,460 mg/kg

LD50 Dermal, guinea pig: >5,000 mg/kg

LC50 Inhalation, rat: >5,100 mg/m³/4H

Draize test, Skin rabbit: 500 uL Severe

Inhalation:

Not available

Irritating:

Diethylene Glycol Ethyl Methyl Ether:

Eye irritating: Moderate (Draize P.I.I=2.5)

Gamma-Butyrolactone:

Eye irritating: (Rabbit OECD 405)

Sensitization:

Not available

Mutagenicity:

Not available

The information shown in SECTION 3, Hazards identification, is based on toxicity profiles of similar materials or on the components present in this material.

12. Ecological information

12.1 Toxicity

Acquatic toxicity: No further relevant information available.

12.2 Persistence and degradability: No further relevant information available.

12.3 Bioaccumulative potential: No further relevant information available.

12.4 Mobility in soil: No further relevant information available.

12.5 Results of PBT and vPvB assessment

PBT: Not applicable.

vPvB: Not applicable.

12.6 Other adverse effects: No further relevant information available.

13. Disposal considerations

13.1 Waste treatment methods

Product

This combustible material may be burned in a chemical incinerator equipped with an afterburner and scrubber. Offer surplus and

non-recyclable solutions to a licensed disposal company. Contact a licensed professional waste disposal service to dispose of this material.

Contaminated packaging

Dispose of as unused product.

14. Transport information

14.1 UN number

ADR/RID: — IMDG: — IATA: —

14.2 UN proper shipping name

ADR/RID: Not dangerous goods

IMDG: Not dangerous goods

IATA: Not dangerous goods

14.3 Transport hazard class(es)

ADR/RID: — IMDG: — IATA: —

14.4 Packaging group

ADR/RID: — IMDG: — IATA: —

14.5 Environmental hazards

ADR/RID: no IMDG Marine pollutant: no IATA: no

14.6 Special precautions for user

no data available

15. Regulatory information

15.1 Safety, health and environmental regulations/legislation specific for the substance or mixture

EU Regulation (EC) No. 1272/2008 (CLP Regulations)

REACH Status: In compliance.

Pre-registration status: All components are listed or exempted.

Annex XIV - List of substances subject to authorization

Substances of very high concern

None of the components are listed.

Annex XVII - Restrictions on the manufacture, placing on the market and use of certain dangerous substances, mixtures and articles

Not applicable.

15.2 Chemical Safety Assessment

No data available

15.3 Other information

US Regulation:

TSCA Section 4(a) Final Test Rules Regulated: Not regulated.

TSCA Section 8(a) Preliminary Assessment Information Rule (PAIR): Not regulated.

TSCA Section 8(a) Inventory Update Rule: All components on TSCA INVENTORY

TSCA Section 8(d) Health and Safety Study Reporting: Not regulated.

TSCA Section 12(b) One-Time Export Notification Regulated: Not regulated.

California Proposition 65: Not regulated.

16. Other information

The data in this Material Safety Data Sheet relates only to the specific material designated herein and does not relate to use in combination with any other material or in any process.

This information is based upon technical information believed to be reliable. It is subject to revision as additional knowledge and experience is gained.

Safety Data Sheet

according to 1907/2006/EC, Article 31

Printing date 01.09.2015 Version: 1 Revision: 01.09.2015



1. Identification of the substance/preparation and of the company/undertaking

1.1 Product identifier

Trade name: PU INK_MG

1.2 Relevant identified uses of the substance or mixture and uses advised against

Material of Use: Industrial applications: Inkjet ink for drop-on-demand digital printing process.

1.3 SDS No.

1.4 Details of the supplier of the safety data sheet

Manufacturer/Supplier: Jetbest Corporation

Address: No. 402-1, Sec. 4, Chun Hwa Road, Hsinchu, 300, Taiwan

Phone: +886-530-5888

e-mail: sales@jetbest.com

Only Representative

Address: Instituto Suizo para el Fomento de la Seguridad-Swissi España S.L.U Calle Lope de Vega 30 Entlo. E-08005 Barcelona

e-mail: reach@swissiesp.com

Emergency telephone number:

Manufacturer: +886-913-323-577 (24hours)

2. Hazards identification

2.1 Classification of the substance or mixture

Classification according to Regulation (EC) No 1272/2008 [CLP]



Danger

Acute Tox. 4 H302 Harmful if swallowed.

Eye Irrit. 2 H318 Causes serious eye irritation.

Skin Irrit.2 ; H315 Causes skin irritation.

STOT SE 3: H336 May cause drowsiness or dizziness.

2.2 Label elements

Labelling according to Regulation (EC) No 1272/2008 [CLP]

Hazard pictograms



Signal word: Danger

Hazard statements:

H302 Harmful if swallowed.

H315 Causes skin irritation.

H318 Causes serious eye irritation.

H336 May cause drowsiness or dizziness.

Precautionary statements:

P261 Avoid breathing dust/ fume/ gas/ mist/ vapours/ spray.

P280 Wear protective gloves/ eye protection/ face protection.

P302+P352 - IF ON SKIN: Wash with plenty of soap and water.

P321 - Specific treatment (see first aid statements on this label).

P305 + P351 + P338 IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.

P362 - Take off contaminated clothing and wash before reuse.

2.3 Other hazards

Results of PBT and vPvB assessment

PBT: Not applicable.

vPvB: Not applicable.

3. Composition/information on ingredients

Chemical characterization: Mixture
Ink Jet printing ink in organic solvents.

Ingredients	CAS-No.	EINECS	EU Pre-registration No.	Percent	Classification	
					67/548/EEC	Regulation (EC) No. 1272/2008 [CLP]
2-Ethoxy Ethyl Ether	112-36-7	203-963-7	Not available for the moment	<25%	Xi; R38	Skin Irrit.2 H315
Diethylene Glycol Ethyl Methyl Ether	1002-67-1	213-690-5	Not available for the moment	40%–70%	—	—
Gamma-Butyrolactone	96-48-0	202-509-5	Not available for the moment	10%–20%	Xi; R22, R41; R67	Acute Tox. 4 H302 Eye Irrit. 2 H318 STOT.SE 3 H336

4. First aid measures

4.1 Description of first aid measures

Eye contact: Rinse thoroughly with plenty of water for at least 15 minutes and consult a physician.
Inhalation: If breathed in, move person into fresh air. If not breathing, give artificial respiration. Consult a physician.
Skin contact: Wash off with soap and plenty of water. Consult a physician.
Ingestion: Do NOT induce vomiting. Never give anything by mouth to an unconscious person. Rinse mouth with water. Consult a physician.

4.2 Most important symptoms and effects, both acute and delayed

Potential acute health effects

Eye contact : No known significant effects or critical hazards.
Inhalation : No known significant effects or critical hazards.
Skin contact : No known significant effects or critical hazards.
Ingestion : No known significant effects or critical hazards.

Over-exposure signs/symptoms

Eye contact : No specific data.
Inhalation : No specific data.
Skin contact : No specific data.
Ingestion : No specific data.

4.3 Indication of any immediate medical attention and special treatment needed

Treat symptomatically.

5. Fire-fighting measures

5.1 Extinguishing media

Suitable extinguishing media

Alcohol-resistant foam, dry chemical, carbon dioxide (CO₂), water-spray.

5.2 Special hazards arising from the substance or mixture

Carbon oxides

5.3 Advice for firefighters

Use breathing apparatus with independent air supply.
Protective suit.

5.4 Further information

Use water spray to cool unopened containers

5.5 NFPA Ratings:

Health: 2 Flammability: 2 Reactivity: 0

Hazard Scale: 0 = Minimal 1 = Slight 2 = Moderate 3 = Serious 4 = Severe

6. Accidental release measures

6.1 Personal precautions, protective equipment and emergency procedures

Use personal protective equipment. Avoid breathing vapors, mist or gas. Ensure adequate ventilation. Remove all sources of ignition. Beware of vapours accumulating to form explosive concentrations. Vapours can accumulate in low areas.

6.2 Environmental precautions

Prevent further leakage or spillage if safe to do so. Do not let product enter drains.

6.3 Methods and materials for containment and cleaning up

Contain spillage, and then collect with an electrically protected vacuum cleaner or by wet-brushing and place in container for disposal according to local regulations (see section 13). Keep in suitable, closed containers for disposal.

6.4 Reference to other sections

For disposal see section 13.

7. Handling and storage

7.1 Handling

Avoid contact with skin and eyes. Avoid inhalation of vapour or mist. Keep away from sources of ignition - No smoking. Take measures to prevent the build up of electrostatic charge.

7.2 Conditions for safe storage, including any incompatibilities

Store in cool place. Keep container tightly closed in a dry and well-ventilated place. Containers which are opened must be carefully resealed and kept upright to prevent leakage.

7.3 Specific end uses

no data available

8. Exposure controls/personal protection

8.1 Control parameters

Components with workplace control parameters

8.2 Exposure controls

Appropriate engineering controls

Handle in accordance with good industrial hygiene and safety practice. Wash hands before breaks and at the end of workday.

Personal protective equipment

Eye/face protection

Face shield and safety glasses Use equipment for eye protection tested and approved under appropriate government standards such as NIOSH (US) or EN 166(EU).

Skin protection

Handle with gloves. Gloves must be inspected prior to use. Use proper glove removal technique (without touching glove's outer surface) to avoid skin contact with this product. Dispose of contaminated gloves after use in accordance with applicable laws and good laboratory practices.

Wash and dry hands.

The selected protective gloves have to satisfy the specifications of EU Directive 89/686/EEC and the standard EN 374 derived from it.

Body Protection

Complete suit protecting against chemicals, The type of protective equipment must be selected according to the concentration and amount of the dangerous substance at the specific workplace.

Respiratory protection

Where risk assessment shows air-purifying respirators are appropriate use a full-face respirator with multi-purpose combination (US) or type ABEK (EN 14387) respirator cartridges as a backup to engineering controls. If the respirator is the sole means of protection, use a full-face supplied air respirator. Use respirators and components tested and approved under appropriate government standards such as NIOSH (US) or CEN (EU).

9. Physical and chemical properties

1	Color	Magenta
2	Odor	Slight odor
3	Boiling point/boiling range of ink	approx. 176 °C or higher
4	Melting Point/Melting Range	No data available
5	Flash point of ink	approx. 70 °C
6	Auto-Ignition Temperature	not below 220 °C
7	Flammability(solid, gas)	Not Applicable
8	Explosive Properties	Lower limits: 2.7 vol% Upper limits: 15.6 vol% (gamma-Butyrolactone)
9	Vapour Pressure	No data available
10	Specific Gravity	0.97 ± 0.01(25°C)
11	Solubility	No data available
12	Water solubility	Easily soluble (Diethylene glycol diethyl ether)
13	Viscosity	4.3 ± 0.5 cps
14	pH	Not applicable
15	Oxidizing properties	No data available
16	Vapor Density	Not Applicable

The Physical and chemical data given in Section 9 are typical values for this product and are not intended to be product specifications.

10. Stability and reactivity

10.1 Reactivity

- no data available
- 10.2 Chemical stability
 - no data available
- 10.3 Possibility of hazardous reactions
 - no data available
- 10.4 Conditions to avoid
 - Heat, flames and sparks.
- 10.5 Incompatible materials
 - Strong oxidizing agents, Strong bases
- 10.6 Hazardous decomposition products
 - Other decomposition products - no data available

11. Toxicological information

11.1 Routes of Overexposure: Eye, skin, inhalation, and oral ingestion

11.2 Health Hazards:

Acute Health Hazards: Overexposure of the eye surface to ink may be mildly irritating. Overexposure of ink contact with the skin may cause irritation and in some people, swelling and redness. Intentional inhalation to ink vapors may result in respiratory tract irritation. Intentional or accidental oral ingestion may cause an upset stomach.

Chronic Health Hazards: No information available

Mutagenicity: No information available

Carcinogenicity: No information available

11.3 Toxicity:

Acute Toxicity Data:

2-Ethoxy Ethyl Ether:

LD50 Oral, rat: 4,970 mg/kg

LD50 Skin, rabbit: 6,700 uL/kg

Draize test, Eye rabbit: 50 mg Moderate

Diethylene Glycol Ethyl Methyl Ether:

LD50 Oral, rat: >6,500mg/kg

LD50 Dermal, rabbit: >7,070mg/kg

Gamma-Butyrolactone:

LD50 Oral, rat: 1,540 mg/kg

LD50 Oral, mouse: 1,460 mg/kg

LD50 Dermal, guinea pig: >5,000 mg/kg

LC50 Inhalation, rat: >5,100 mg/m³/4H

Draize test, Skin rabbit: 500 uL Severe

Inhalation:

Not available

Irritating:

Diethylene Glycol Ethyl Methyl Ether:

Eye irritating: Moderate (Draize P.I.I=2.5)

Gamma-Butyrolactone:

Eye irritating: (Rabbit OECD 405)

Sensitization:

Not available

Mutagenicity:

Not available

The information shown in SECTION 3, Hazards identification, is based on toxicity profiles of similar materials or on the components present in this material.

12. Ecological information

12.1 Toxicity

Acquatic toxicity: No further relevant information available.

12.2 Persistence and degradability: No further relevant information available.

12.3 Bioaccumulative potential: No further relevant information available.

12.4 Mobility in soil: No further relevant information available.

12.5 Results of PBT and vPvB assessment

PBT: Not applicable.

vPvB: Not applicable.

12.6 Other adverse effects: No further relevant information available.

13. Disposal considerations

13.1 Waste treatment methods

Product

This combustible material may be burned in a chemical incinerator equipped with an afterburner and scrubber. Offer surplus and

non-recyclable solutions to a licensed disposal company. Contact a licensed professional waste disposal service to dispose of this material.

Contaminated packaging

Dispose of as unused product.

14. Transport information

14.1 UN number

ADR/RID: — IMDG: — IATA: —

14.2 UN proper shipping name

ADR/RID: Not dangerous goods

IMDG: Not dangerous goods

IATA: Not dangerous goods

14.3 Transport hazard class(es)

ADR/RID: — IMDG: — IATA: —

14.4 Packaging group

ADR/RID: — IMDG: — IATA: —

14.5 Environmental hazards

ADR/RID: no IMDG Marine pollutant: no IATA: no

14.6 Special precautions for user

no data available

15. Regulatory information

15.1 Safety, health and environmental regulations/legislation specific for the substance or mixture

EU Regulation (EC) No. 1272/2008 (CLP Regulations)

REACH Status: In compliance.

Pre-registration status: All components are listed or exempted.

Annex XIV - List of substances subject to authorization

Substances of very high concern

None of the components are listed.

Annex XVII - Restrictions on the manufacture, placing on the market and use of certain dangerous substances, mixtures and articles

Not applicable.

15.2 Chemical Safety Assessment

No data available

15.3 Other information

US Regulation:

TSCA Section 4(a) Final Test Rules Regulated: Not regulated.

TSCA Section 8(a) Preliminary Assessment Information Rule (PAIR): Not regulated.

TSCA Section 8(a) Inventory Update Rule: All components on TSCA INVENTORY

TSCA Section 8(d) Health and Safety Study Reporting: Not regulated.

TSCA Section 12(b) One-Time Export Notification Regulated: Not regulated.

California Proposition 65: Not regulated.

16. Other information

The data in this Material Safety Data Sheet relates only to the specific material designated herein and does not relate to use in combination with any other material or in any process.

This information is based upon technical information believed to be reliable. It is subject to revision as additional knowledge and experience is gained.

Safety Data Sheet

according to 1907/2006/EC, Article 31

Printing date 01.09.2015 Version: 1 Revision: 01.09.2015



1. Identification of the substance/preparation and of the company/undertaking

1.1 Product identifier

Trade name: PU INK_YE

1.2 Relevant identified uses of the substance or mixture and uses advised against

Material of Use: Industrial applications: Inkjet ink for drop-on-demand digital printing process.

1.3 SDS No.

1.4 Details of the supplier of the safety data sheet

Manufacturer/Supplier: Jetbest Corporation

Address: No. 402-1, Sec. 4, Chun Hwa Road, Hsinchu, 300, Taiwan

Phone: +886-530-5888

e-mail: sales@jetbest.com

Only Representative

Address: Instituto Suizo para el Fomento de la Seguridad-Swissi España S.L.U Calle Lope de Vega 30 Entlo. E-08005 Barcelona

e-mail: reach@swissiesp.com

Emergency telephone number:

Manufacturer: +886-913-323-577 (24hours)

2. Hazards identification

2.1 Classification of the substance or mixture

Classification according to Regulation (EC) No 1272/2008 [CLP]



Danger

Acute Tox. 4 H302 Harmful if swallowed.

Acute tox 4: H312 Harmful in contact with skin.

Eye Irrit. 2 H318 Causes serious eye irritation.

Skin Irrit.2 ; H315 Causes skin irritation.

Acute tox 4: H332 Harmful if inhaled.

STOT SE 3: H336 May cause drowsiness or dizziness.

2.2 Label elements

Labelling according to Regulation (EC) No 1272/2008 [CLP]

Hazard pictograms



Signal word: Danger

Hazard statements:

H302 Harmful if swallowed.

H312 Harmful in contact with skin.

H315 Causes skin irritation.

H318 Causes serious eye irritation.

H332 Harmful if inhaled.

H336 May cause drowsiness or dizziness.

Precautionary statements:

P261 Avoid breathing dust/ fume/ gas/ mist/ vapours/ spray.

P280 Wear protective gloves/ eye protection/ face protection.

P302+P352 - IF ON SKIN: Wash with plenty of soap and water.

P321 - Specific treatment (see first aid statements on this label).

P305 + P351 + P338 IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.

P332+P313 - If skin irritation occurs: Get medical advice/attention.

P362 - Take off contaminated clothing and wash before reuse.

2.3 Other hazards

Results of PBT and vPvB assessment

PBT: Not applicable.

vPvB: Not applicable.

3. Composition/information on ingredients

Chemical characterization: Mixture

Ink Jet printing ink in organic solvents.

Ingredients	CAS-No.	EINECS	EU Pre-registration No.	Percent	Classification	
					67/548/EEC	Regulation (EC) No. 1272/2008 [CLP]
2-Ethoxy Ethyl Ether	112-36-7	203-963-7	Not available for the moment	<5%	Xi; R38	Skin Irrit.2 H315
Diethylene Glycol Ethyl Methyl Ether	1002-67-1	213-690-5	Not available for the moment	40%–70%	—	—
Gamma-Butyrolactone	96-48-0	202-509-5	Not available for the moment	10%–20%	Xi; R22, R41; R67	Acute Tox. 4 H302 Eye Irrit. 2 H318 STOT.SE 3 H336
Ethylene Glycol Monobutyl Ether Acetate	112-07-2	203-933-3	Not available for the moment	<20%	Xn; R20/21	Acute tox 4: H312, Acute tox 4: H332

4. First aid measures

4.1 Description of first aid measures

Eye contact: Rinse thoroughly with plenty of water for at least 15 minutes and consult a physician.

Inhalation: If breathed in, move person into fresh air. If not breathing, give artificial respiration. Consult a physician.

Skin contact: Wash off with soap and plenty of water. Consult a physician.

Ingestion: Do NOT induce vomiting. Never give anything by mouth to an unconscious person. Rinse mouth with water. Consult a physician.

4.2 Most important symptoms and effects, both acute and delayed

Potential acute health effects

Eye contact : No known significant effects or critical hazards.

Inhalation : No known significant effects or critical hazards.

Skin contact : No known significant effects or critical hazards.

Ingestion : No known significant effects or critical hazards.

Over-exposure signs/symptoms

Eye contact : No specific data.

Inhalation : No specific data.

Skin contact : No specific data.

Ingestion : No specific data.

4.3 Indication of any immediate medical attention and special treatment needed

Treat symptomatically.

5. Fire-fighting measures

5.1 Extinguishing media

Suitable extinguishing media

Alcohol-resistant foam, dry chemical, carbon dioxide (CO₂), water-spray.

5.2 Special hazards arising from the substance or mixture

Carbon oxides

5.3 Advice for firefighters

Use breathing apparatus with independent air supply.

Protective suit.

5.4 Further information

Use water spray to cool unopened containers

5.5 NFPA Ratings:

Health: 2 Flammability: 2 Reactivity: 0

Hazard Scale: 0 = Minimal 1 = Slight 2 = Moderate 3 = Serious 4 = Severe

6. Accidental release measures

6.1 Personal precautions, protective equipment and emergency procedures

Use personal protective equipment. Avoid breathing vapors, mist or gas. Ensure adequate ventilation. Remove all sources of ignition. Beware of vapours accumulating to form explosive concentrations. Vapours can accumulate in low areas.

6.2 Environmental precautions

Prevent further leakage or spillage if safe to do so. Do not let product enter drains.

6.3 Methods and materials for containment and cleaning up

Contain spillage, and then collect with an electrically protected vacuum cleaner or by wet-brushing and place in container for disposal according to local regulations (see section 13). Keep in suitable, closed containers for disposal.

6.4 Reference to other sections
For disposal see section 13.

7. Handling and storage

7.1 Handling

Avoid contact with skin and eyes. Avoid inhalation of vapour or mist. Keep away from sources of ignition - No smoking. Take measures to prevent the build up of electrostatic charge.

7.2 Conditions for safe storage, including any incompatibilities

Store in cool place. Keep container tightly closed in a dry and well-ventilated place. Containers which are opened must be carefully resealed and kept upright to prevent leakage.

7.3 Specific end uses

no data available

8. Exposure controls/personal protection

8.1 Control parameters

Components with workplace control parameters

8.2 Exposure controls

Appropriate engineering controls

Handle in accordance with good industrial hygiene and safety practice. Wash hands before breaks and at the end of workday.

Personal protective equipment

Eye/face protection

Face shield and safety glasses Use equipment for eye protection tested and approved under appropriate government standards such as NIOSH (US) or EN 166(EU).

Skin protection

Handle with gloves. Gloves must be inspected prior to use. Use proper glove removal technique (without touching glove's outer surface) to avoid skin contact with this product. Dispose of contaminated gloves after use in accordance with applicable laws and good laboratory practices.

Wash and dry hands.

The selected protective gloves have to satisfy the specifications of EU Directive 89/686/EEC and the standard EN 374 derived from it.

Body Protection

Complete suit protecting against chemicals, The type of protective equipment must be selected according to the concentration and amount of the dangerous substance at the specific workplace.

Respiratory protection

Where risk assessment shows air-purifying respirators are appropriate use a full-face respirator with multi-purpose combination (US) or type ABEK (EN 14387) respirator cartridges as a backup to engineering controls. If the respirator is the sole means of protection, use a full-face supplied air respirator. Use respirators and components tested and approved under appropriate government standards such as NIOSH (US) or CEN (EU).

9. Physical and chemical properties

1	Color	Yellow
2	Odor	Slight odor
3	Boiling point/boiling range of ink	approx. 176 °C or higher
4	Melting Point/Melting Range	No data available
5	Flash point of ink	approx. 70 °C
6	Auto-Ignition Temperature	not below 220 °C
7	Flammability(solid, gas)	Not Applicable
8	Explosive Properties	Lower limits: 2.7 vol% Upper limits: 15.6 vol% (gamma-Butyrolactone) Lower limits: 0.88 vol% Upper limits: 8.54 vol% (Ethylene Glycol Monobutyl Ether Acetate)
9	Vapour Pressure	No data available
10	Specific Gravity	0.97 ± 0.01(25°C)
11	Solubility	No data available
12	Water solubility	Easily soluble (Diethylene glycol diethyl ether)
13	Viscosity	4.3 ± 0.5 cps
14	pH	Not applicable

15	Oxidizing properties	No data available
16	Vapor Density	Not Applicable

The Physical and chemical data given in Section 9 are typical values for this product and are not intended to be product specifications.

10. Stability and reactivity

10.1 Reactivity

no data available

10.2 Chemical stability

no data available

10.3 Possibility of hazardous reactions

no data available

10.4 Conditions to avoid

Heat, flames and sparks.

10.5 Incompatible materials

Strong oxidizing agents, Strong bases

10.6 Hazardous decomposition products

Other decomposition products - no data available

11. Toxicological information

11.1 Routes of Overexposure: Eye, skin, inhalation, and oral ingestion

11.2 Health Hazards:

Acute Health Hazards: Overexposure of the eye surface to ink may be mildly irritating. Overexposure of ink contact with the skin may cause irritation and in some people, swelling and redness. Intentional inhalation to ink vapors may result in respiratory tract irritation. Intentional or accidental oral ingestion may cause an upset stomach.

Chronic Health Hazards: No information available

Mutagenicity: No information available

Carcinogenicity: No information available

11.3 Toxicity:

Acute Toxicity Data:

2-Ethoxy Ethyl Ether:

LD50 Oral, rat: 4,970 mg/kg

LD50 Skin, rabbit: 6,700 uL/kg

Draize test, Eye rabbit: 50 mg Moderate

Diethylene Glycol Ethyl Methyl Ether:

LD50 Oral, rat: >6,500mg/kg

LD50 Dermal, rabbit: >7,070mg/kg

Gamma-Butyrolactone:

LD50 Oral, rat: 1,540 mg/kg

LD50 Oral, mouse: 1,460 mg/kg

LD50 Dermal, guinea pig: >5,000 mg/kg

LC50 Inhalation, rat: >5,100 mg/m³/4H

Draize test, Skin rabbit: 500 uL Severe

Ethylene Glycol Monobutyl Ether Acetate:

LD50 Oral, rat: 2,400 mg/kg

LD50 Oral, mouse: 3,200 mg/kg

LD50 Skin, rabbit: 1,500 mg/kg

Draize test, Eye rabbit: 500 mg/24H Mild

Inhalation:

Not available

Irritating:

Ethylene Glycol Monobutyl Ether Acetate:

Eye irritating: 500mg/24H Mild (Rabbit OECD405)

Skin irritating: 500mg/24H Mild (open@ Rabbit OECD404)

Diethylene Glycol Ethyl Methyl Ether:

Eye irritating: Moderate (Draize P.I.I=2.5)

Gamma-Butyrolactone:

Eye irritating: (Rabbit OECD 405)

Sensitization:

Not available

Mutagenicity:

Not available

The information shown in SECTION 3, Hazards identification, is based on toxicity profiles of similar materials or on the

components present in this material.

12. Ecological information

12.1 Toxicity

Acquatic toxicity: No further relevant information available.

12.2 Persistence and degradability: No further relevant information available.

12.3 Bioaccumulative potential: No further relevant information available.

12.4 Mobility in soil: No further relevant information available.

12.5 Results of PBT and vPvB assessment

PBT: Not applicable.

vPvB: Not applicable.

12.6 Other adverse effects: No further relevant information available.

13. Disposal considerations

13.1 Waste treatment methods

Product

This combustible material may be burned in a chemical incinerator equipped with an afterburner and scrubber. Offer surplus and non-recyclable solutions to a licensed disposal company. Contact a licensed professional waste disposal service to dispose of this material.

Contaminated packaging

Dispose of as unused product.

14. Transport information

14.1 UN number

ADR/RID: — IMDG: — IATA: —

14.2 UN proper shipping name

ADR/RID: Not dangerous goods

IMDG: Not dangerous goods

IATA: Not dangerous goods

14.3 Transport hazard class(es)

ADR/RID: — IMDG: — IATA: —

14.4 Packaging group

ADR/RID: — IMDG: — IATA: —

14.5 Environmental hazards

ADR/RID: no IMDG Marine pollutant: no IATA: no

14.6 Special precautions for user

no data available

15. Regulatory information

15.1 Safety, health and environmental regulations/legislation specific for the substance or mixture

EU Regulation (EC) No. 1272/2008 (CLP Regulations)

REACH Status: In compliance.

Pre-registration status: All components are listed or exempted.

Annex XIV - List of substances subject to authorization

Substances of very high concern

None of the components are listed.

Annex XVII - Restrictions on the manufacture, placing on the market and use of certain dangerous substances, mixtures and articles

Not applicable.

15.2 Chemical Safety Assessment

No data available

15.3 Other information

US Regulation:

TSCA Section 4(a) Final Test Rules Regulated: Not regulated.

TSCA Section 8(a) Preliminary Assessment Information Rule (PAIR): Not regulated.

TSCA Section 8(a) Inventory Update Rule: All components on TSCA INVENTORY

TSCA Section 8(d) Health and Safety Study Reporting: Not regulated.

TSCA Section 12(b) One-Time Export Notification Regulated: Not regulated.

California Proposition 65: Not regulated.

16. Other information

The data in this Material Safety Data Sheet relates only to the specific material designated herein and does not relate to use in combination with any other material or in any process.

This information is based upon technical information believed to be reliable. It is subject to revision as additional knowledge and experience is gained.

Safety Data Sheet

according to 1907/2006/EC, Article 31

Printing date 01.09.2015 Version: 1 Revision: 01.09.2015



1. Identification of the substance/preparation and of the company/undertaking

1.1 Product identifier

Trade name: PU INK_BK

1.2 Relevant identified uses of the substance or mixture and uses advised against

Material of Use: Industrial applications: Inkjet ink for drop-on-demand digital printing process.

1.3 SDS No.

1.4 Details of the supplier of the safety data sheet

Manufacturer/Supplier: Jetbest Corporation

Address: No. 402-1, Sec. 4, Chun Hwa Road, Hsinchu, 300, Taiwan

Phone: +886-530-5888

e-mail: sales@jetbest.com

Only Representative

Address: Instituto Suizo para el Fomento de la Seguridad-Swissi España S.L.U Calle Lope de Vega 30 Entlo. E-08005 Barcelona

e-mail: reach@swissiesp.com

Emergency telephone number:

Manufacturer: +886-913-323-577 (24hours)

2. Hazards identification

2.1 Classification of the substance or mixture

Classification according to Regulation (EC) No 1272/2008 [CLP]



Danger

Acute Tox. 4 H302 Harmful if swallowed.

Eye Irrit. 2 H318 Causes serious eye irritation.

Skin Irrit. 2 ; H315 Causes skin irritation.

STOT SE 3: H336 May cause drowsiness or dizziness.

2.2 Label elements

Labelling according to Regulation (EC) No 1272/2008 [CLP]

Hazard pictograms



Signal word: Danger

Hazard statements:

H302 Harmful if swallowed.

H315 Causes skin irritation.

H318 Causes serious eye irritation.

H336 May cause drowsiness or dizziness.

Precautionary statements:

P261 Avoid breathing dust/ fume/ gas/ mist/ vapours/ spray.

P280 Wear protective gloves/ eye protection/ face protection.

P302+P352 - IF ON SKIN: Wash with plenty of soap and water.

P321 - Specific treatment (see first aid statements on this label).

P305 + P351 + P338 IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.

P362 - Take off contaminated clothing and wash before reuse.

2.3 Other hazards

Results of PBT and vPvB assessment

PBT: Not applicable.

vPvB: Not applicable.

3. Composition/information on ingredients

Chemical characterization: Mixture
Ink Jet printing ink in organic solvents.

Ingredients	CAS-No.	EINECS	EU Pre-registration No.	Percent	Classification	
					67/548/EEC	Regulation (EC) No. 1272/2008 [CLP]
2-Ethoxy Ethyl Ether	112-36-7	203-963-7	Not available for the moment	<25%	Xi; R38	Skin Irrit.2 H315
Diethylene Glycol Ethyl Methyl Ether	1002-67-1	213-690-5	Not available for the moment	40%–70%	—	—
Gamma-Butyrolactone	96-48-0	202-509-5	Not available for the moment	10%–20%	Xi; R22, R41; R67	Acute Tox. 4 H302 Eye Irrit. 2 H318 STOT.SE 3 H336

4. First aid measures

4.1 Description of first aid measures

Eye contact: Rinse thoroughly with plenty of water for at least 15 minutes and consult a physician.
Inhalation: If breathed in, move person into fresh air. If not breathing, give artificial respiration. Consult a physician.
Skin contact: Wash off with soap and plenty of water. Consult a physician.
Ingestion: Do NOT induce vomiting. Never give anything by mouth to an unconscious person. Rinse mouth with water. Consult a physician.

4.2 Most important symptoms and effects, both acute and delayed

Potential acute health effects

Eye contact : No known significant effects or critical hazards.
Inhalation : No known significant effects or critical hazards.
Skin contact : No known significant effects or critical hazards.
Ingestion : No known significant effects or critical hazards.

Over-exposure signs/symptoms

Eye contact : No specific data.
Inhalation : No specific data.
Skin contact : No specific data.
Ingestion : No specific data.

4.3 Indication of any immediate medical attention and special treatment needed

Treat symptomatically.

5. Fire-fighting measures

5.1 Extinguishing media

Suitable extinguishing media

Alcohol-resistant foam, dry chemical, carbon dioxide (CO₂), water-spray.

5.2 Special hazards arising from the substance or mixture

Carbon oxides

5.3 Advice for firefighters

Use breathing apparatus with independent air supply.
Protective suit.

5.4 Further information

Use water spray to cool unopened containers

5.5 NFPA Ratings:

Health: 2 Flammability: 2 Reactivity: 0

Hazard Scale: 0 = Minimal 1 = Slight 2 = Moderate 3 = Serious 4 = Severe

6. Accidental release measures

6.1 Personal precautions, protective equipment and emergency procedures

Use personal protective equipment. Avoid breathing vapors, mist or gas. Ensure adequate ventilation. Remove all sources of ignition. Beware of vapours accumulating to form explosive concentrations. Vapours can accumulate in low areas.

6.2 Environmental precautions

Prevent further leakage or spillage if safe to do so. Do not let product enter drains.

6.3 Methods and materials for containment and cleaning up

Contain spillage, and then collect with an electrically protected vacuum cleaner or by wet-brushing and place in container for disposal according to local regulations (see section 13). Keep in suitable, closed containers for disposal.

6.4 Reference to other sections

For disposal see section 13.

7. Handling and storage

7.1 Handling

Avoid contact with skin and eyes. Avoid inhalation of vapour or mist. Keep away from sources of ignition - No smoking. Take measures to prevent the build up of electrostatic charge.

7.2 Conditions for safe storage, including any incompatibilities

Store in cool place. Keep container tightly closed in a dry and well-ventilated place. Containers which are opened must be carefully resealed and kept upright to prevent leakage.

7.3 Specific end uses

no data available

8. Exposure controls/personal protection

8.1 Control parameters

Components with workplace control parameters

8.2 Exposure controls

Appropriate engineering controls

Handle in accordance with good industrial hygiene and safety practice. Wash hands before breaks and at the end of workday.

Personal protective equipment

Eye/face protection

Face shield and safety glasses Use equipment for eye protection tested and approved under appropriate government standards such as NIOSH (US) or EN 166(EU).

Skin protection

Handle with gloves. Gloves must be inspected prior to use. Use proper glove removal technique (without touching glove's outer surface) to avoid skin contact with this product. Dispose of contaminated gloves after use in accordance with applicable laws and good laboratory practices.

Wash and dry hands.

The selected protective gloves have to satisfy the specifications of EU Directive 89/686/EEC and the standard EN 374 derived from it.

Body Protection

Complete suit protecting against chemicals, The type of protective equipment must be selected according to the concentration and amount of the dangerous substance at the specific workplace.

Respiratory protection

Where risk assessment shows air-purifying respirators are appropriate use a full-face respirator with multi-purpose combination (US) or type ABEK (EN 14387) respirator cartridges as a backup to engineering controls. If the respirator is the sole means of protection, use a full-face supplied air respirator. Use respirators and components tested and approved under appropriate government standards such as NIOSH (US) or CEN (EU).

9. Physical and chemical properties

1	Color	Black
2	Odor	Slight odor
3	Boiling point/boiling range of ink	approx. 176 °C or higher
4	Melting Point/Melting Range	No data available
5	Flash point of ink	approx. 70 °C
6	Auto-Ignition Temperature	not below 220 °C
7	Flammability(solid, gas)	Not Applicable
8	Explosive Properties	Lower limits: 2.7 vol% Upper limits: 15.6 vol% (gamma-Butyrolactone) Lower limits: 0.88 vol% Upper limits: 8.54 vol% (Ethylene Glycol Monobutyl Ether Acetate)
9	Vapour Pressure	No data available
10	Specific Gravity	0.97 ± 0.01(25°C)
11	Solubility	No data available
12	Water solubility	Easily soluble (Diethylene glycol diethyl ether)
13	Viscosity	4.3 ± 0.5 cps
14	pH	Not applicable
15	Oxidizing properties	No data available
16	Vapor Density	Not Applicable

The Physical and chemical data given in Section 9 are typical values for this product and are not intended to be product specifications.

10. Stability and reactivity

10.1 Reactivity

no data available

10.2 Chemical stability

no data available

10.3 Possibility of hazardous reactions

no data available

10.4 Conditions to avoid

Heat, flames and sparks.

10.5 Incompatible materials

Strong oxidizing agents, Strong bases

10.6 Hazardous decomposition products

Other decomposition products - no data available

11. Toxicological information

11.1 Routes of Overexposure: Eye, skin, inhalation, and oral ingestion

11.2 Health Hazards:

Acute Health Hazards: Overexposure of the eye surface to ink may be mildly irritating. Overexposure of ink contact with the skin may cause irritation and in some people, swelling and redness. Intentional inhalation to ink vapors may result in respiratory tract irritation. Intentional or accidental oral ingestion may cause an upset stomach.

Chronic Health Hazards: No information available

Mutagenicity: No information available

Carcinogenicity: With excessive exposure, carbon black has been listed as a possible human carcinogen. However, as engineered within this ink cartridge, emissions to air of carbon black during normal printing use have not been found. IARC, the International Agency for Research on Cancer, has found printing inks to be not classifiable as human carcinogens.

11.3 Toxicity:

Acute Toxicity Data:

2-Ethoxy Ethyl Ether:

LD50 Oral, rat: 4,970 mg/kg

LD50 Skin, rabbit: 6,700 uL/kg

Draize test, Eye rabbit: 50 mg Moderate

Diethylene Glycol Ethyl Methyl Ether:

LD50 Oral, rat: >6,500mg/kg

LD50 Dermal, rabbit: >7,070mg/kg

Gamma-Butyrolactone:

LD50 Oral, rat: 1,540 mg/kg

LD50 Oral, mouse: 1,460 mg/kg

LD50 Dermal, guinea pig: >5,000 mg/kg

LC50 Inhalation, rat: >5,100 mg/m³/4H

Draize test, Skin rabbit: 500 uL Severe

Inhalation:

Not available

Irritating:

Diethylene Glycol Ethyl Methyl Ether:

Eye irritating: Moderate (Draize P.I.I=2.5)

Gamma-Butyrolactone:

Eye irritating: (Rabbit OECD 405)

Sensitization:

Not available

Mutagenicity:

Not available

The information shown in SECTION 3, Hazards identification, is based on toxicity profiles of similar materials or on the components present in this material.

12. Ecological information

12.1 Toxicity

Acquatic toxicity: No further relevant information available.

12.2 Persistence and degradability: No further relevant information available.

12.3 Bioaccumulative potential: No further relevant information available.

12.4 Mobility in soil: No further relevant information available.

12.5 Results of PBT and vPvB assessment

PBT: Not applicable.

vPvB: Not applicable.

12.6 Other adverse effects: No further relevant information available.

13. Disposal considerations

13.1 Waste treatment methods

Product

This combustible material may be burned in a chemical incinerator equipped with an afterburner and scrubber. Offer surplus and non-recyclable solutions to a licensed disposal company. Contact a licensed professional waste disposal service to dispose of this material.

Contaminated packaging

Dispose of as unused product.

14. Transport information

14.1 UN number

ADR/RID: — IMDG: — IATA: —

14.2 UN proper shipping name

ADR/RID: Not dangerous goods

IMDG: Not dangerous goods

IATA: Not dangerous goods

14.3 Transport hazard class(es)

ADR/RID: — IMDG: — IATA: —

14.4 Packaging group

ADR/RID: — IMDG: — IATA: —

14.5 Environmental hazards

ADR/RID: no IMDG Marine pollutant: no IATA: no

14.6 Special precautions for user

no data available

15. Regulatory information

15.1 Safety, health and environmental regulations/legislation specific for the substance or mixture

EU Regulation (EC) No. 1272/2008 (CLP Regulations)

REACH Status: In compliance.

Pre-registration status: All components are listed or exempted.

Annex XIV - List of substances subject to authorization

Substances of very high concern

None of the components are listed.

Annex XVII - Restrictions on the manufacture, placing on the market and use of certain dangerous substances, mixtures and articles

Not applicable.

15.2 Chemical Safety Assessment

No data available

15.3 Other information

US Regulation:

TSCA Section 4(a) Final Test Rules Regulated: Not regulated.

TSCA Section 8(a) Preliminary Assessment Information Rule (PAIR): Not regulated.

TSCA Section 8(a) Inventory Update Rule: All components on TSCA INVENTORY

TSCA Section 8(d) Health and Safety Study Reporting: Not regulated.

TSCA Section 12(b) One-Time Export Notification Regulated: Not regulated.

California Proposition 65: Not regulated.

16. Other information

The data in this Material Safety Data Sheet relates only to the specific material designated herein and does not relate to use in combination with any other material or in any process.

This information is based upon technical information believed to be reliable. It is subject to revision as additional knowledge and experience is gained.